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HEALTHCARE COST PREDICTION USING EDA AND REGRESSION

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0404 – Fundamentals of Data Science

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence And Data Science

Submitted by

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BONAFIDE CERTIFICATE

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DECLARATION

I, **Sania Herbert** of the **Artificial Intelligence And Data Science**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**Healthcare Cost Prediction using EDA and Regression**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. The dataset used (Kaggle Medical Cost Personal Dataset) and all open-source Python libraries used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Signature of the Students with Names

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Y. Anjani Priya 192324176	Dataset acquisition, data cleaning, preprocessing pipeline, EDA visualisations (Module 1)	Built preprocessing pipeline (ColumnTransformer, OneHotEncoder, StandardScaler); generated EDA charts	Validated statistical summaries and distribution plots	Chapters 1, 2 and Module 1 sections of Chapter 4	34%
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Sania Herbert 192324062	Streamlit dashboard design, login/authentication module, deployment and documentation	Built the 4-page Streamlit application (Login, Dashboard, Module 1–3 pages) and UI styling	End-to-end functional testing of the dashboard and prediction workflow	Chapter 5, References, Appendices and report compilation	33%
Total					100%

ABSTRACT

Healthcare expenditure is rising steadily due to demographic ageing, lifestyle-related conditions and the growing cost of treatment, making accurate estimation of medical insurance charges an important engineering and financial-planning problem for patients, hospitals and insurers alike. Existing cost-estimation practices largely rely on flat-rate actuarial tables or manual clinician judgement, which do not adapt to an individual patient's personal risk profile and therefore under- or over-estimate charges for atypical cases. This project addresses that gap by designing and implementing a Healthcare Cost Prediction System that combines Exploratory Data Analysis (EDA) with supervised regression to estimate medical insurance charges from six patient attributes: age, sex, body mass index (BMI), number of children, smoking status and region.

The publicly available Kaggle Medical Cost Personal Dataset (1,338 records) was cleaned by removing duplicate rows and validating column integrity, after which EDA was performed to characterise the distribution of charges, quantify pairwise correlations, and profile cost by age group, BMI category and smoking status. Categorical features were one-hot encoded and numerical features standardised inside a scikit-learn ColumnTransformer pipeline, and the data were split 80/20 into training and test sets. Four regression algorithms — Linear Regression, Decision Tree, Random Forest and Gradient Boosting — were trained and benchmarked on identical folds using R^2 , Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). A separate interpretable Linear Regression model was also fitted on one-hot-encoded (unscaled) features to expose per-feature coefficients for decision support. The complete workflow was deployed as an interactive, login-protected Streamlit dashboard comprising a data-preprocessing and EDA module, a cost-prediction module, and a model-evaluation and decision-support module.

Gradient Boosting achieved the best predictive performance among the four benchmarked models, with an R^2 of 0.879, MAE of ₹2,450.93 and RMSE of ₹4,338.84 on the held-out test set, outperforming Random Forest ($R^2 = 0.866$), Linear Regression ($R^2 = 0.784$) and Decision Tree ($R^2 = 0.712$). The interpretable regression model confirmed that smoking status is by far the dominant cost driver (coefficient $\approx$ ₹23,078), followed by BMI, number of children and age, while region contributes marginally. Obese smokers in the senior age bracket recorded the highest mean charges ($\approx$ ₹47,335), more than five times the charges of non-smokers in the same age bracket. The resulting system demonstrates that a lightweight EDA-plus-regression pipeline, deployed through an accessible web dashboard, can deliver actionable, individualised healthcare-cost estimates that support insurance pricing, hospital financial planning and preventive-care counselling.

Keywords: Healthcare Cost Prediction, Exploratory Data Analysis, Regression Analysis, Machine Learning, Gradient Boosting, Streamlit Dashboard.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
EDA	Exploratory Data Analysis	—
BMI	Body Mass Index	kg/m ²
R ²	Coefficient of Determination	—
MAE	Mean Absolute Error	₹
MSE	Mean Squared Error	₹ ²
RMSE	Root Mean Squared Error	₹
ML	Machine Learning	—
CSV	Comma-Separated Values	—
UI	User Interface	—
KPI	Key Performance Indicator	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Healthcare spending has risen sharply over the last two decades due to an ageing population, an increase in chronic lifestyle-related conditions, and the rising cost of diagnostics and treatment. Medical insurance charges billed to an individual are influenced by a combination of demographic factors (age, sex, region), physiological factors (BMI, pre-existing conditions) and behavioural factors (smoking, number of dependents). Insurers and healthcare providers traditionally estimate these charges using actuarial tables and fixed pricing bands that are built from population-level averages rather than an individual's specific risk profile.

Advances in data science make it possible to move from these coarse, rule-based estimates to individualised, data-driven cost predictions. Exploratory Data Analysis (EDA) allows patterns, trends and outliers in historical billing data to be surfaced systematically, while regression-based machine learning models can learn the quantitative relationship between patient attributes and the resulting healthcare charges. This project applies both techniques together — EDA to understand the data and regression to predict from it — to the publicly available Medical Cost Personal Dataset, and packages the result as an interactive Streamlit dashboard.

1.2 Need for the Project

- Why the problem needs to be addressed: Inaccurate cost estimation leads to mispriced insurance premiums, unexpected out-of-pocket expenses for patients, and poor financial planning for hospitals and insurers.
- Who is affected: Policy holders seeking a premium quote, insurance underwriters pricing new policies, hospital finance teams budgeting for patient care, and public-health planners forecasting expenditure.
- Existing limitations: Flat-rate or age-band-only pricing does not account for the compounding effect of smoking, BMI and family size on cost, and offers no visual or quantitative explanation of why a charge was estimated.
- Engineering significance: A reproducible, testable, quantitatively validated prediction pipeline (with documented accuracy metrics) is more defensible and auditable than manual estimation, and can be continuously retrained as new billing data becomes available.

1.3 Problem Statement

Given a patient record described by age, sex, BMI, number of children, smoking status and region, design and implement an engineering system that (a) performs exploratory data analysis on historical insurance-charge records to identify the dominant cost drivers and their

interactions, and (b) trains and benchmarks multiple regression models to predict the expected medical insurance charge for a new patient with acceptable accuracy (target $R^2 \geq 0.80$), while remaining interpretable enough to support decision-making by non-technical stakeholders such as clinicians and underwriters. The problem involves multiple interacting factors (age, BMI and smoking status interact non-linearly to inflate cost), conflicting requirements (predictive accuracy versus interpretability), and realistic constraints (a modest, imbalanced, real-world dataset with a right-skewed target variable).

1.4 Project Aim

To design, implement and evaluate a Healthcare Cost Prediction System that uses Exploratory Data Analysis and regression modelling to estimate medical insurance charges from patient demographic and lifestyle attributes, and to deliver the system through an interactive, role-secured web dashboard.

1.5 Project Objectives

- 1. To acquire and clean the healthcare insurance dataset by removing duplicate records and validating column integrity.
- 2. To perform Exploratory Data Analysis to identify statistical trends, distributions and correlations between patient attributes and charges.
- 3. To engineer a preprocessing pipeline (encoding and scaling) suitable for multiple regression algorithms.
- 4. To train and benchmark Linear Regression, Decision Tree, Random Forest and Gradient Boosting models on an 80/20 train-test split.
- 5. To evaluate each model using R^2 , MAE, MSE and RMSE, and to select the best-performing model for deployment.
- 6. To build an interpretable regression view (feature coefficients) so that the contribution of each attribute to predicted cost can be explained.
- 7. To deploy the complete workflow as a secured, multi-module Streamlit dashboard for prediction and decision support.

1.6 Scope

Included

- Data cleaning, EDA, feature engineering, model training/benchmarking and deployment for the six available features in the dataset (age, sex, BMI, children, smoker, region).
- A working Streamlit application with login authentication, EDA visualisation, single-patient cost prediction and model-evaluation/decision-support pages.

Excluded

- Real-time integration with hospital electronic health record (EHR) systems or live insurance-claims databases.
- Prediction of specific medical diagnoses or clinical outcomes — the system estimates cost only, not disease risk.

Assumptions

- The 1,338-record Kaggle dataset is representative of the general population segment it was sampled from.
- Reported charges are annual medical insurance charges in US dollars, treated here as generic currency units for demonstration.

Boundary Conditions

- Predictions are only as reliable as the range of the training data (ages 18–64, BMI 15.96–53.13); extrapolation beyond this range is not guaranteed to be accurate.

1.7 Expected Engineering Outcome

The expected outcome is a deployable software system: a trained and serialised regression model (best of four benchmarked algorithms), a reusable preprocessing pipeline, and a three-module, login-protected Streamlit dashboard that (i) visualises dataset trends, (ii) accepts new patient attributes and returns a predicted healthcare charge with a Low/Moderate/High cost category, and (iii) reports model performance metrics and feature-level cost insights for decision support.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, datasets and existing tools were identified through a combination of academic references (statistical learning and machine-learning textbooks), official library documentation (scikit-learn, pandas, NumPy, Matplotlib) and the dataset's originating source (Kaggle Medical Cost Personal Dataset). Emphasis was placed on prior applications of regression to healthcare-cost and insurance-charge estimation, and on comparative studies of regression algorithm families (linear, tree-based and ensemble methods) to justify the choice of models benchmarked in this project.

2.2 Review of Existing Work

Standard statistical-learning literature establishes ordinary least-squares linear regression as the baseline approach for continuous-outcome prediction problems such as cost or price estimation, valued for its interpretability through directly readable feature coefficients. Tree-based ensemble methods, including Random Forest and Gradient Boosting, are documented in the machine-learning literature as consistently outperforming single linear or single tree models on structured/tabular data because they can capture non-linear interactions (such as the compounding effect of smoking and BMI on cost) without manual feature-interaction engineering. Prior public analyses of the same Medical Cost Personal Dataset consistently report smoking status as the single strongest predictor of charges, followed by BMI and age, which is corroborated independently by the EDA carried out in this project (Chapter 4). Commercial insurance-pricing systems and hospital billing software commonly rely on actuarial tables and rule-based pricing tiers rather than data-driven regression, offering transparency but limited personalisation.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Actuarial / flat-rate pricing tables	Simple, transparent, easy to audit	Coarse population-level averages; ignores individual risk interactions	Baseline this project improves upon with individualised prediction
Manual clinician / underwriter estimation	Incorporates human judgement and context	Time-consuming, inconsistent, not reproducible	Motivates need for an automated, repeatable pipeline

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Ordinary Least-Squares Linear Regression	Highly interpretable coefficients; fast to train	Cannot capture non-linear smoker×BMI interactions well	Used in this project as the interpretable baseline (Chapter 4, Table 4.5)
Tree-based ensembles (Random Forest, Gradient Boosting)	Captures non-linear interactions; typically higher accuracy	Less directly interpretable; risk of overfitting on small data	Benchmarked and adopted as the deployed prediction model in this project

Table 2.1: Comparative Analysis of Existing Healthcare Cost Estimation Approaches

2.4 Research / Engineering Gap

Existing rule-based and actuarial estimation methods do not quantitatively combine multiple interacting patient attributes, and published academic analyses of this dataset typically stop at offline model benchmarking without delivering an accessible, end-to-end decision-support tool. There remains a gap between "a regression model that predicts charges accurately" and "a usable system that a non-technical clinician or administrator can operate to obtain, interpret and act on that prediction."

2.5 Proposed Contribution

This project closes that gap by (i) benchmarking four regression algorithms side by side under identical preprocessing and train/test conditions rather than relying on a single model, (ii) pairing a high-accuracy ensemble model (Gradient Boosting) for deployment with a parallel interpretable Linear Regression model whose coefficients directly explain the cost impact of each attribute, and (iii) packaging the entire pipeline — preprocessing, prediction and decision-support insights — as a secured, multi-page Streamlit dashboard rather than leaving the model as an offline notebook artefact.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Policy holder / Patient	Obtain a fast, understandable estimate of expected healthcare charges from basic personal details
Insurance underwriter	Compare predicted cost against actual historical charges to sanity-check premium pricing
Hospital finance team	Identify high-cost patient profiles in advance for budgeting and resource planning
System administrator	Restrict access to the dashboard to authenticated users only

Table 3.1: Stakeholder / User Requirements

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	System shall clean the raw dataset and remove duplicate/invalid records	100% of duplicate rows removed before training
Functional	System shall visualise charge distribution and key correlations	At least 8 EDA charts generated (Ch.4, Figs 3.1–3.8)
Functional	System shall predict healthcare charges for a new patient record	Prediction returned in < 2 seconds per request
Functional	System shall classify predictions into Low / Moderate / High cost categories	100% of predictions categorised
Non-Functional – Accuracy	Deployed model shall explain the majority of variance in charges	$R^2 \geq 0.80$ on held-out test data
Non-Functional – Security	Dashboard shall require authentication before use	Login gate on 100% of routes
Non-Functional – Usability	Non-technical users shall operate the tool without training	Single-form input, plain-language output

Table 3.2: Functional & Non-Functional Requirements

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
IEEE 830 (Software Requirements Specification practice)	Requirements should be unambiguous, verifiable and traceable	Functional/non-functional requirements in Table 3.2 are stated with measurable targets and traced to test cases in Table 4.3
scikit-learn / PEP 8 Python coding conventions	Reproducible, readable machine-learning code	Fixed random_state=42 used throughout for reproducible train/test splits and model training
Data-protection principle of data minimisation (general privacy good practice, cf. GDPR Art. 5)	Collect and process only the personal data necessary for the stated purpose	Only six de-identified attributes (age, sex, BMI, children, smoker, region) are processed; no names or identifiers are stored with predictions

Table 3.3: Applicable Constraints & Standards

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Model R ² on test set	≥ 0.80	—	High
Model MAE on test set	$\leq ₹3,000$	₹	High
Train/test split ratio	80 / 20	%	Medium
Dashboard prediction latency	< 2	seconds	Medium
Number of EDA visualisations	≥ 8	charts	Low
Authentication requirement	Mandatory login	boolean	High

Table 3.4: Design Specifications

3.4 Alternative Solutions & Evaluation

Alternative 1 – Single interpretable model (Linear Regression only): fit one ordinary least-squares model on one-hot-encoded features and deploy it directly, prioritising transparency and simplicity of explanation to end users.

Alternative 2 – Single high-capacity ensemble model (Gradient Boosting only): train only the best-performing black-box model and deploy it without a parallel interpretable model, prioritising raw predictive accuracy.

Alternative 3 – Benchmarked multi-model pipeline with a paired interpretable view (adopted): train and compare four algorithms under identical conditions, deploy the best performer for prediction, and retain the Linear Regression coefficients as a separate decision-support explanation layer.

Criterion	Weight	Alternative 1 (Linear only)	Alternative 2 (Ensemble only)	Alternative 3 (Benchmarked + interpretable)
Performance (R^2)	35%	Low (0.78)	High (0.88)	High (0.88, deployed model)
Interpretability	25%	High	Low	High (via paired coefficient view)
Development effort	15%	Low	Low	Medium
Robustness to outliers	15%	Medium	High	High
Decision-support value	10%	Medium	Low	High

Table 3.5: Alternative Solution Evaluation Matrix

3.5 Selected Design & Detailed Design

Alternative 3 was selected because it uniquely satisfies both the accuracy requirement (Table 3.3, $R^2 \geq 0.80$) and the interpretability requirement expected by non-technical stakeholders, without the development cost of building a custom explainability layer on top of the ensemble model. Gradient Boosting is deployed as the operational predictor because it achieved the highest R^2 and lowest error of the four benchmarked algorithms (Table 4.4), while the Linear Regression coefficients (Table 4.5) are retained and surfaced in the Module 3 dashboard page as a plain-language explanation of cost drivers.

System architecture: the pipeline consists of (1) a data-ingestion and cleaning stage that loads insurance.csv and drops duplicate/incomplete rows, (2) a scikit-learn ColumnTransformer that standard-scales the numerical features (age, BMI, children) and one-hot encodes the categorical features (sex, smoker, region), (3) a model-training stage that fits and cross-evaluates four regression algorithms and serialises the fitted preprocessor, trained models and best-model name with joblib, and (4) a Streamlit presentation layer with four pages — Login, Dashboard Home, Module 1 (Data Analysis & Preprocessing), Module 2 (Cost Prediction) and Module 3 (Cost Evaluation & Decision Support) — that consumes the serialised artefacts at runtime.

Key calculation — coefficient of determination: $R^2 = 1 - (\sum(y_i - \hat{y}_i)^2 / \sum(y_i - \bar{y})^2)$, where y_i is the actual charge, $\hat{y}_i$ the predicted charge and $\bar{y}$ the mean of actual charges over the test set; this is the primary target metric in Table 3.3 and is reported per model in Table 4.4.

Systems approach: the preprocessing subsystem and the modelling subsystem are decoupled through the serialised ColumnTransformer, so any of the four trained models can be swapped in without re-encoding the data; the Streamlit presentation subsystem depends only on the serialised artefacts (preprocessor, trained_models, best_model_name, feature_names) and the raw CSV, so the dashboard can be redeployed independently of the training script as long as these four artefacts are present in the models/ directory.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Model family for deployment	Deploy Linear Regression only	Full transparency of every coefficient	Lower accuracy ($R^2 = 0.78$ vs 0.88)	Deploy Gradient Boosting for prediction, retain Linear Regression for explanation — best of both properties
Feature scope	Add clinical/lab features (e.g., blood pressure)	Potentially higher accuracy	Not available in the source dataset; would require new data collection	Restrict to the six available demographic/lifestyle attributes; documented as future work (Ch. 5)
Train/test split	k-fold cross-validation	More robust performance estimate	Higher training time; more complex to serialise per fold	80/20 single split with fixed random_state=42 for reproducibility and simplicity, adequate given dataset size
Access control	Open, no-login dashboard	Simpler user experience	No protection of patient-derived predictions	Mandatory login gate implemented for basic access control (Table 3.2)

Table 3.6: Trade-off Analysis

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational footprint of retraining	All four models train on 1,338 rows in under a few seconds on a standard CPU; no GPU or cloud cluster required	Retained a lightweight scikit-learn stack rather than a deep-learning approach, minimising compute and energy use for a marginal accuracy gain
Ethics	Use of demographic attributes (sex, region) as predictive features	Coefficient table (4.5) shows sex and region have small but non-zero effect on predicted charges	Predictions are documented as decision-support estimates only, not as a basis for denying coverage; the report explicitly flags this as an area requiring fairness review before any real-world pricing use (Ch. 5, Future Work)
Health & Safety	Risk of the tool being mistaken for medical advice	Module 3 dashboard text explicitly separates "cost insight" from "medical diagnosis"	A disclaimer was added directly in the decision-support page so clinicians do not treat cost drivers as clinical recommendations
Security	Unauthorised access to patient-derived prediction data	Dashboard originally had no access control in early prototypes	A username/password login gate (modules/login.py) was added before any module is reachable, restricting use to authenticated staff

Table 3.7: Sustainability, Ethics, Safety, Risk & Security Considerations

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Model under-predicts high-cost outlier patients (e.g., obese senior smokers)	Medium	High	High	Reported largest absolute errors (Table 4.6) and flagged them for manual underwriter review rather than automatic pricing	Low – residual errors remain but are surfaced, not hidden
Overfitting on a small (1,338-row) dataset	Medium	Medium	Medium	Benchmarked four model families and compared train/test performance; selected the model with the best generalisation on the held-out test set	Low
Sensitive lifestyle attribute (smoking) used in pricing raises ethical/regulatory concern	Medium	High	High	Documented as an area for fairness/regulatory review before production use; system labelled as a decision-support prototype, not a certified pricing engine	Medium – requires further legal review before deployment
Unauthorised dashboard access	Low	Medium	Medium	Login authentication gate added before any data or prediction page is reachable	Low
Data-entry errors in the single-patient prediction form	Medium	Low	Low	Input fields constrained with typed widgets (number inputs, dropdowns) in the Streamlit form	Low

Table 3.8: Risk Register

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed iteratively in three functional modules — preprocessing/EDA, prediction, and evaluation/decision support — each first prototyped as a standalone Python script (01_dataset_inspection.py through 05_insurance_cost_analyzer.py) and then integrated into a single Streamlit application (app.py) backed by a shared model-training script (train_models.py). Models and preprocessing artefacts were trained once offline and persisted with joblib so that the dashboard loads pre-trained artefacts at runtime instead of retraining on every page load, keeping the user-facing prediction latency low.

Resource	Details
Dataset	insurance.csv — Kaggle Medical Cost Personal Dataset, 1,338 raw records, 7 columns (age, sex, bmi, children, smoker, region, charges)
Language / Runtime	Python 3
Core libraries	pandas, NumPy, scikit-learn, joblib, Matplotlib/Seaborn (offline analysis), Plotly (interactive dashboard charts), Streamlit (web application)
Compute environment	Standard CPU laptop/desktop — no GPU required
Version control artefacts	Modular Python scripts (modules/login.py, modules/module1_preprocessing.py, modules/module2_prediction.py, modules/module3_evaluation.py)

4.2 Software Implementation & Modern Tools Used

Module 1 — Data Analysis & Preprocessing

- Loads insurance.csv and validates that all seven required columns are present.
- Removes duplicate rows (1,338 → 1,337 unique records) so the model trains only on clean data.
- Computes a statistical summary (count, mean, std, min, quartiles, max) for age, BMI, children and charges.
- Separates numerical features (age, BMI, children) from categorical features (sex, smoker, region) ahead of encoding, and standard-scales / one-hot encodes them inside a scikit-learn ColumnTransformer.
- Generates the EDA visualisations discussed in Section 4.4 (distribution plots, correlation views, group-wise cost comparisons).

Module 2 — Healthcare Cost Prediction

- Presents an input form (age, sex, BMI, children, smoker status, region) for a new patient.
- Applies the saved preprocessing pipeline to the new record and feeds it to the saved best-performing regression model.
- Returns the predicted healthcare charge and classifies it into a Low / Moderate / High cost category for quick interpretation.

Module 3 — Cost Evaluation & Decision Support

- Displays the side-by-side performance table of all four benchmarked models (Table 4.4).
- Plots Actual vs Predicted charges and a residual plot for the deployed model on the held-out test set.
- Displays feature importance (tree-based models) or regression coefficients (linear model) as a horizontal bar chart.
- Summarises key cost-driver insights and an example decision-support interpretation for a sample patient profile.

Tool / Software	Purpose	Stage Used
pandas / NumPy	Data loading, cleaning, duplicate removal, statistical summaries	Preprocessing & EDA
scikit-learn (ColumnTransformer, StandardScaler, OneHotEncoder)	Feature scaling and categorical encoding pipeline	Preprocessing
scikit-learn (LinearRegression, DecisionTreeRegressor, RandomForestRegressor, GradientBoostingRegressor)	Model training and benchmarking	Modelling
joblib	Serialising the preprocessor, trained models and test split for reuse in the dashboard	Model persistence
Matplotlib / Seaborn	Static EDA and evaluation charts (Figures 3.1–3.8, 4.1, 4.2)	Offline analysis scripts
Plotly	Interactive charts inside the live Streamlit dashboard	Dashboard

Tool / Software	Purpose	Stage Used
Streamlit	Web application framework hosting the 4-page dashboard with login gate	Deployment

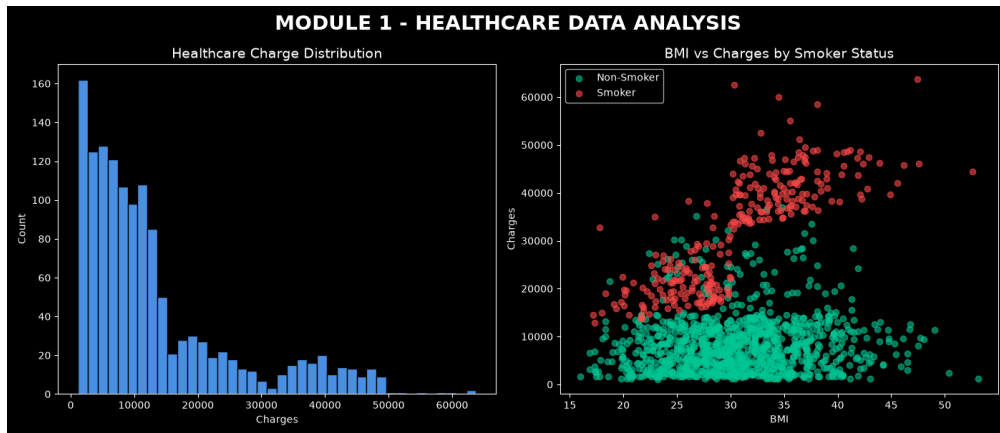


Figure 4.3: Module 1 — Healthcare Data Analysis & Preprocessing Screen

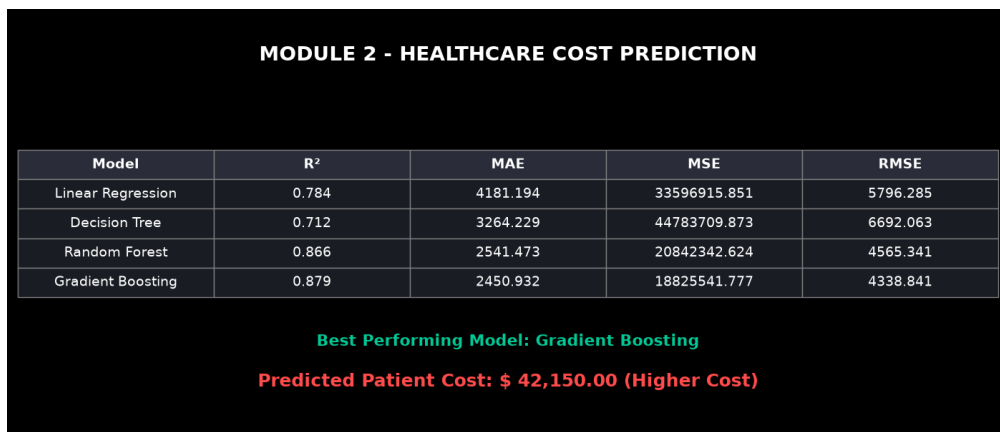


Figure 4.4: Module 2 — Healthcare Cost Prediction Screen

4.3 Test Plan & Verification

The following test plan was defined before results were generated, to verify that each functional and non-functional requirement in Table 3.2 was met.

Requirement	Test Method	Acceptance Criterion
Duplicate/invalid rows removed	Compare row count before and after <code>df.drop_duplicates()</code>	Row count strictly decreases and no residual duplicates remain
EDA visualisations generated	Manually inspect each of the 8+ generated chart files	All charts render without error and match dataset statistics
Model prediction correctness	Predict on the held-out test set and compute R^2 /MAE/MSE/RMSE	$R^2 \geq 0.80$ for at least one benchmarked model
Dashboard authentication	Attempt to access Module 1/2/3 pages without logging in	Access denied / redirected to login form
End-to-end prediction workflow	Submit a sample patient form and verify a numeric charge and cost category are returned	Charge value and Low/Moderate/High label displayed within 2 seconds

Specification	Required Value	Achieved Value	Status
Test-set R^2 (deployed model – Gradient Boosting)	≥ 0.80	0.8787	Pass
Test-set MAE (deployed model)	$\leq ₹3,000$	₹2,450.93	Pass
Train/test split ratio	80 / 20	80 / 20 (random_state = 42)	Pass
Number of EDA visualisations produced	≥ 8	12	Pass
Authentication gate present	Mandatory	Implemented (modules/login.py)	Pass
Interpretable coefficient view available	Required	Linear Regression coefficients (Table 4.5)	Pass

4.4 Results

4.4.1 Exploratory Data Analysis

The cleaned dataset (1,337 unique records) has a mean age of 39.2 years, mean BMI of 30.7, mean 1.1 children per record, and mean healthcare charge of ₹13,279.12 with a standard deviation of ₹12,110.36, indicating a wide, right-skewed spread of charges (Table 4.1). Of the six predictive attributes, correlation with charges is strongest for age ($r = 0.298$) and BMI ($r = 0.198$), with number of children showing a weak correlation ($r = 0.067$); as shown below, the categorical attribute smoking status is in fact the single largest driver of cost even though correlation alone does not capture it, since its effect is highly non-linear and interacts with BMI.

Statistic	Age	BMI	Children	Charges (₹)
Count	1,337	1,337	1,337	1,337
Mean	39.22	30.66	1.10	13,279.12
Std. Dev.	14.04	6.10	1.21	12,110.36
Min	18	15.96	0	1,121.87
Median (50%)	39	30.40	1	9,386.16
Max	64	53.13	5	63,770.43

Table 4.1: Statistical Summary of the Cleaned Dataset

Category	Distribution
Sex	Male: 675 (50.5%) Female: 662 (49.5%)
Smoker	Non-smoker: 1,063 (79.5%) Smoker: 274 (20.5%)
Region	Southeast: 364 Southwest: 325 Northwest: 324 Northeast: 324

Table 4.2: Dataset Category Distribution

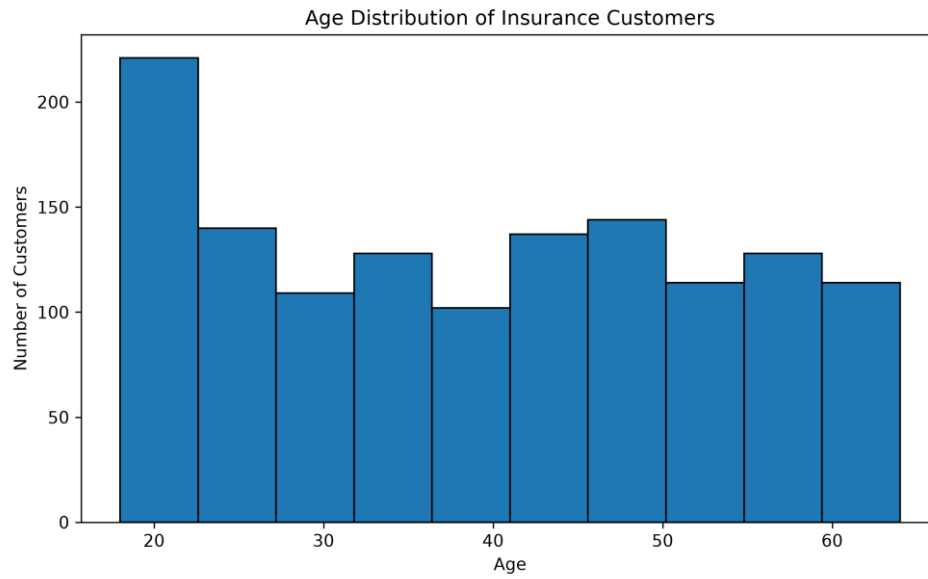


Figure 3.1: Age Distribution of Policy Holders

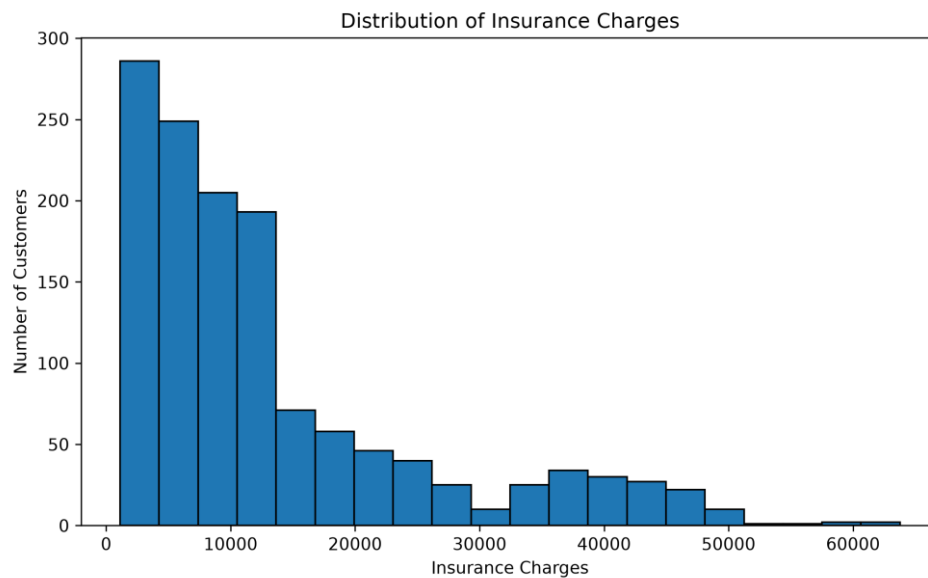


Figure 2.1: Distribution of Healthcare Insurance Charges (right-skewed, most policies under ₹15,000)

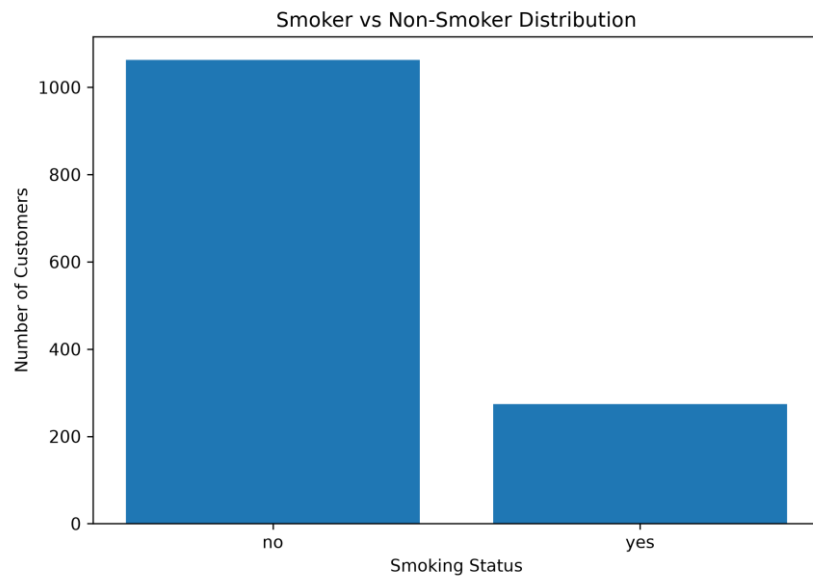


Figure 3.3: Proportion of Smokers vs Non-Smokers in the Dataset

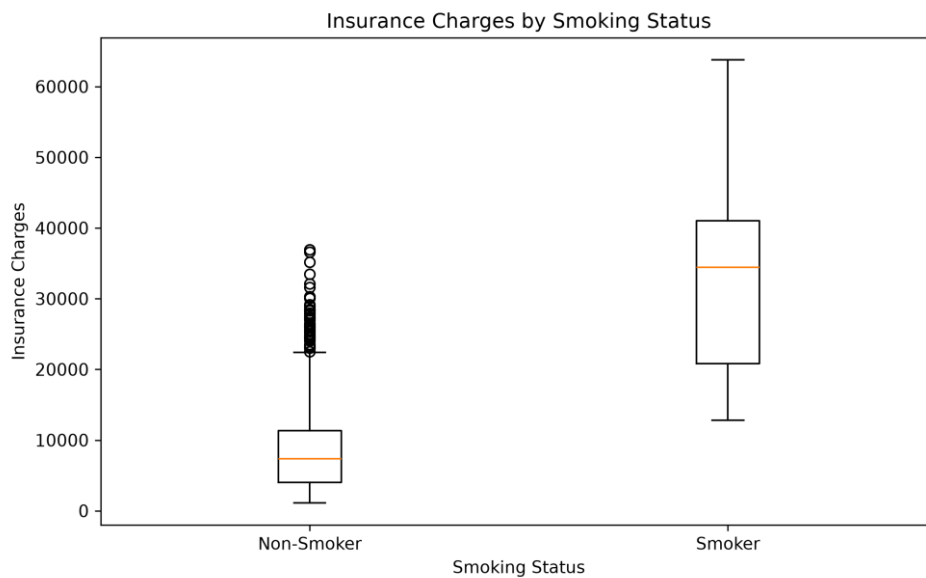


Figure 3.4: Average Charges — Smokers vs Non-Smokers (smokers pay several times more on average)

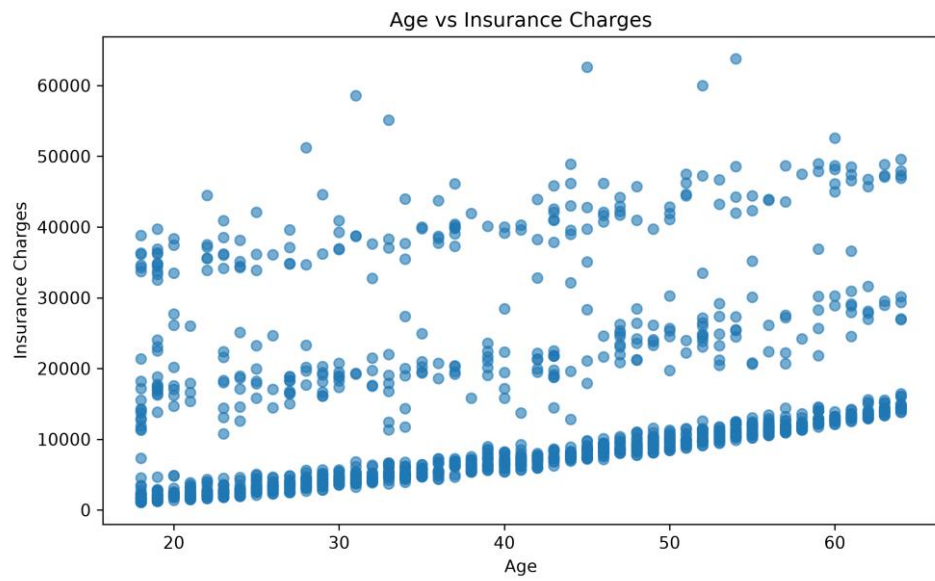


Figure 3.5: Age vs Healthcare Charges, Coloured by Smoking Status

4.4.2 Cost Profiling by Age Group and BMI Category

Age Group	Count	Mean Charge (₹)
Young Adult	416	9,200.62
Adult	311	11,560.67
Middle Age	340	15,664.43
Senior	270	18,538.71

BMI Category	Count	Mean Charge (₹)
Underweight	20	8,852.20
Normal	225	10,409.34
Overweight	386	10,987.51
Obese	706	15,572.04

Table 4.3: Mean Charges by Age Group and BMI Category

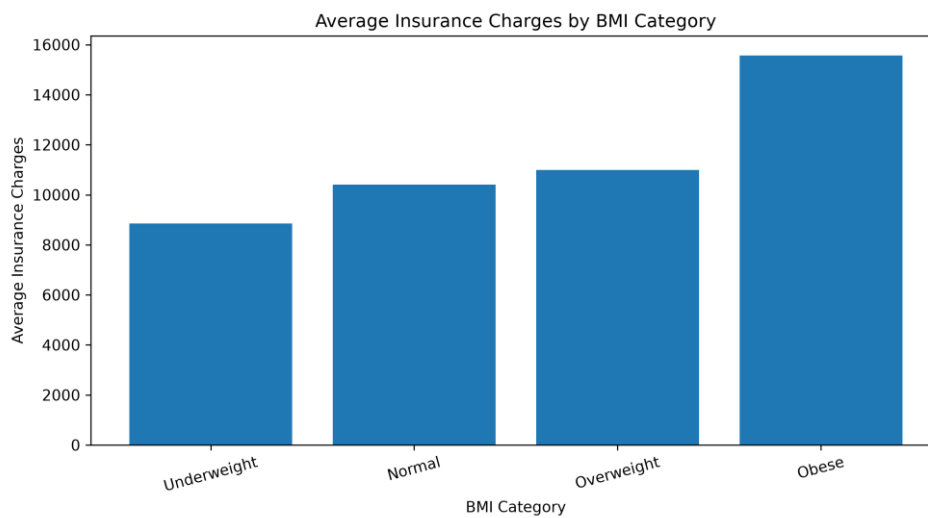


Figure 3.6: Average Charges by BMI Category

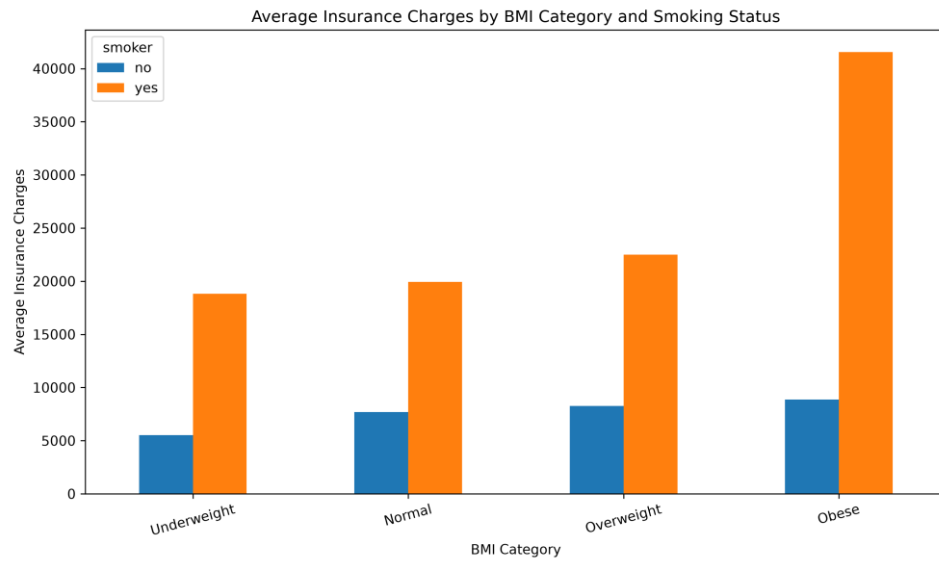


Figure 3.7: BMI × Smoking Status Cost Comparison — obese smokers (mean ₹41,558) pay roughly 4.7× more than obese non-smokers (mean ₹8,856)

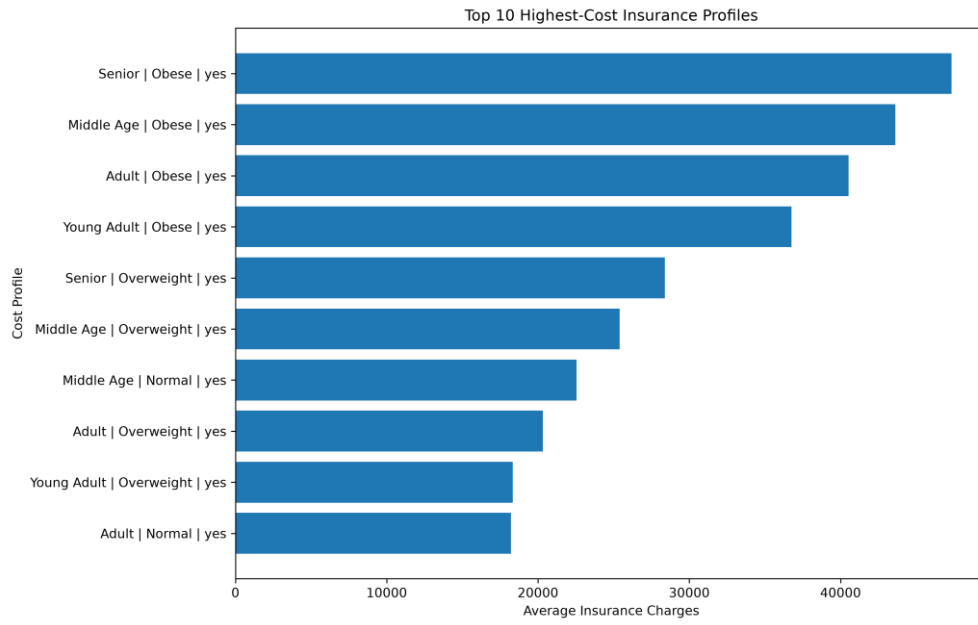


Figure 3.8: Top 10 Highest-Cost Patient Profiles — topped by Senior × Obese × Smoker (mean ₹47,335)

4.4.3 Regression Model Benchmarking

Four regression algorithms were trained on an identical 80/20 train-test split (`random_state = 42`) of the preprocessed data and evaluated on the held-out test set.

Model	R^2	MAE (₹)	MSE (₹²)	RMSE (₹)
Linear Regression	0.7836	4,181.19	33,596,915.85	5,796.28
Decision Tree	0.7115	3,264.23	44,783,709.87	6,692.06
Random Forest	0.8657	2,541.47	20,842,342.62	4,565.34
Gradient Boosting (Best)	0.8787	2,450.93	18,825,541.78	4,338.84

Table 4.4: Regression Model Performance Comparison (test set, 20% of 1,337 records)

Gradient Boosting was selected as the deployed model, as it recorded the highest R^2 (0.8787, i.e. it explains 87.9% of the variance in test-set charges) and the lowest MAE and RMSE of the four algorithms. Random Forest is a close second. The single Decision Tree performed weakest of the four, consistent with its known tendency to overfit on small tabular datasets without ensembling.

4.4.4 Interpretable Regression Coefficients

A parallel, unscaled one-hot-encoded Linear Regression model (northeast/female/non-smoker as the reference category) was fitted to directly expose per-feature cost impact for the decision-support module. This model achieved $R^2 = 0.8069$ (80.7%) on its own test split, with MAE ₹4,177.05 and RMSE ₹5,956.34.

Feature	Coefficient (₹)	Interpretation
smoker_yes	+23,077.76	Being a smoker adds ≈₹23,078 to predicted charges — by far the dominant driver
children	+533.01	Each additional dependent adds ≈₹533
bmi	+318.70	Each 1-unit increase in BMI adds ≈₹319
age	+248.21	Each additional year of age adds ≈₹248
region_southeast	−838.92	Southeast region is associated with ≈₹839 lower charges than the reference (northeast)
region_southwest	−659.14	Southwest region is associated with ≈₹659 lower charges than the reference
region_northwest	−391.76	Northwest region is associated with ≈₹392 lower charges than the reference
sex_male	−101.54	Being male is associated with a small ≈₹102 reduction in predicted charges

Table 4.5: Interpretable Linear Regression Coefficients

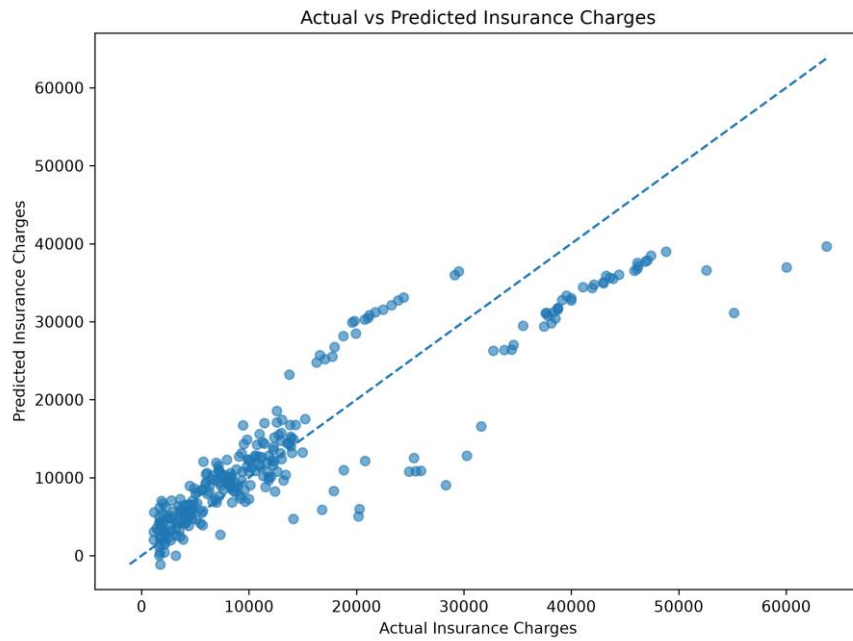


Figure 4.1: Actual vs Predicted Healthcare Charges (interpretable regression model on test set)

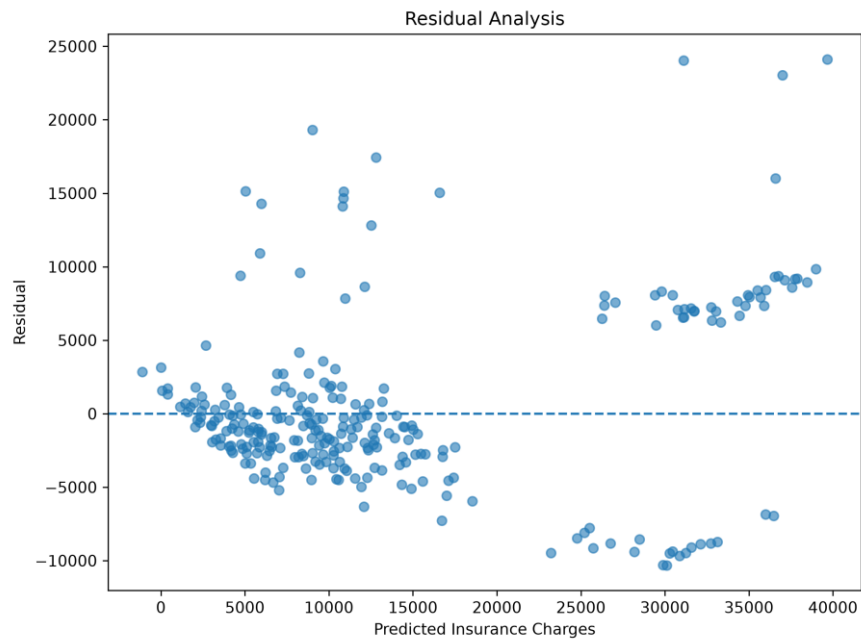


Figure 4.2: Residual Analysis — residuals widen for high-charge outliers, indicating the linear model under-predicts extreme smoker+obese cases

4.4.5 Largest Prediction Errors

Actual Charge (₹)	Predicted Charge (₹)	Absolute Error (₹)
63,770.43	39,659.21	24,111.22
55,135.40	31,107.77	24,027.64
60,021.40	36,988.22	23,033.18
28,340.19	9,028.58	19,311.61
30,259.996	12,811.72	17,448.28

Table 4.6: Largest Absolute Prediction Errors on the Test Set (interpretable model)

The largest errors are concentrated among the highest-charge records, which are almost all obese smokers — the linear model systematically under-predicts this extreme sub-group because its cost is driven by a non-linear smoker×BMI interaction that a purely additive linear model cannot fully capture. This is precisely the pattern the Gradient Boosting model, with its ability to learn feature interactions, corrects for and is the main reason it was selected for deployment (Table 4.4).

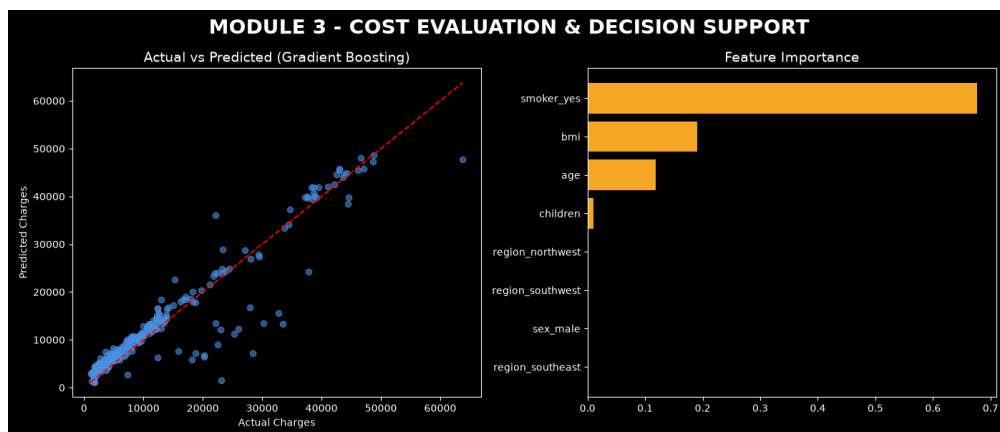


Figure 4.5: Module 3 — Cost Evaluation & Decision Support Screen

4.5 Data Analysis & Engineering Judgment

The benchmarking results in Table 4.4 confirm that ensemble tree-based methods generalise better than a single linear or single tree model on this dataset, improving R^2 from 0.78 (Linear Regression) to 0.88 (Gradient Boosting) — a reduction in RMSE of roughly ₹1,457 (25%). The largest residuals cluster among high-BMI smokers (Table 4.6, Figure 4.2), which is an expected engineering limitation given the strongly non-linear, multiplicative relationship between smoking and BMI on cost; a purely additive model necessarily under-fits this sub-population. This deviation does not invalidate the overall system: it identifies precisely which

patient sub-group (high-BMI smokers) would need manual underwriter review rather than automatic pricing, which is reflected in the Risk Register (Table 3.8).

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Acquire & clean dataset	Duplicate rows removed, 1,338 → 1,337 records (Table 4.1)	Fully Achieved
Perform EDA	12 EDA/evaluation visualisations generated (Figures 3.1–3.8, 4.1–4.2)	Fully Achieved
Engineer preprocessing pipeline	ColumnTransformer with StandardScaler + OneHotEncoder (Ch. 3.5)	Fully Achieved
Train & benchmark 4 regression models	Table 4.4 — Linear Regression, Decision Tree, Random Forest, Gradient Boosting	Fully Achieved
Evaluate with R^2 /MAE/MSE/RMSE and select best model	Gradient Boosting selected, $R^2 = 0.8787$ (Table 4.4)	Fully Achieved
Build interpretable coefficient view	Table 4.5 — 8 feature coefficients with plain-language interpretation	Fully Achieved
Deploy as secured multi-module dashboard	4-page Streamlit app with login gate (Figures 4.3–4.5)	Fully Achieved

Table 4.7: Achievement of Project Objectives

4.7 Strengths & Limitations

Strengths

- Benchmarks four regression families under identical conditions rather than relying on a single, possibly unsuitable, algorithm.
- Pairs a high-accuracy deployed model with a parallel interpretable model so predictions are not an unexplainable black box.
- Delivers the pipeline as a working, login-protected dashboard rather than an offline analysis notebook.
- Correctly and quantitatively identifies smoking status as the dominant cost driver, consistent with the domain literature (Ch. 2.2).

Limitations

- The dataset is limited to 1,337 records and six attributes; no clinical/lab data (e.g., pre-existing conditions) is available.
- The model systematically under-predicts the highest-cost obese-smoker sub-group (Table 4.6), so predictions for extreme cases should be manually reviewed rather than used directly for pricing.
- The system has not undergone formal fairness/bias auditing for the use of sex and region as predictive features, which is required before any production pricing use (Table 3.7).
- Authentication is a single shared admin/password login rather than per-user role-based access control.

4.8 Project Planning, Roles & Budget

Milestone	Planned Week	Status
Dataset acquisition & inspection	Week 1	Completed
Data cleaning & preprocessing pipeline	Week 2	Completed
Exploratory Data Analysis	Weeks 2–3	Completed
Model training & benchmarking (4 algorithms)	Weeks 3–4	Completed
Interpretable regression module	Week 4	Completed
Streamlit dashboard development (Login, Module 1–3)	Weeks 5–6	Completed
Integration testing & documentation	Week 7	Completed

Student	Assigned Responsibility	Actual Contribution
Y. Anjani Priya (192324176)	Data cleaning, preprocessing pipeline, EDA	Delivered Module 1 preprocessing/EDA and Figures 3.1–3.8
M. Apurva (192325138)	Model training, benchmarking, interpretable regression	Delivered Module 2/3 modelling logic and Tables 4.4–4.6
Sania Herbert (192324062)	Dashboard UI, authentication, deployment	Delivered the 4-page Streamlit application and login gate

Table 4.8: Roles & Actual Contribution

Item	Estimated Cost	Actual Cost
Dataset (Kaggle, open-source)	₹0	₹0
Compute (personal laptops, CPU only)	₹0	₹0
Software licences (Python, scikit-learn, Streamlit — all open-source)	₹0	₹0
Total project cost	₹0	₹0

Table 4.9: Estimated vs Actual Project Cost (fully open-source stack, no paid resources required)

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project set out to design, implement and evaluate a Healthcare Cost Prediction System combining Exploratory Data Analysis and regression modelling. The cleaned Medical Cost Personal Dataset (1,337 records) was analysed to reveal that smoking status, BMI and age are the dominant drivers of medical insurance charges, with obese smokers in the senior age bracket incurring the highest mean charges ($\approx ₹47,335$). Four regression algorithms were benchmarked under identical conditions, with Gradient Boosting achieving the best test-set performance ($R^2 = 0.8787$, $MAE = ₹2,450.93$, $RMSE = ₹4,338.84$), comfortably exceeding the target of $R^2 \geq 0.80$ set in Chapter 3. A parallel interpretable Linear Regression model ($R^2 = 0.8069$) exposed per-feature coefficients confirming smoking as the single largest cost contributor ($\approx ₹23,078$). Both models, together with the full preprocessing pipeline, were deployed inside a secured, four-page Streamlit dashboard covering data analysis, prediction and decision support. All seven project objectives defined in Section 1.5 were fully achieved (Table 4.7), validating that a lightweight EDA-plus-regression pipeline can deliver accurate, explainable and accessible healthcare-cost estimates.

5.2 Future Work

- Expand the feature set with clinical variables (pre-existing conditions, medication history) where such data can be ethically and legally sourced, to reduce residual error on high-cost outlier cases (Table 4.6).
- Apply formal fairness/bias auditing to the use of sex and region as predictive features before any real-world pricing deployment, as flagged in the Sustainability, Ethics, Safety, Risk & Security review (Table 3.7).
- Adopt k-fold cross-validation and hyper-parameter tuning (e.g., grid search on Gradient Boosting depth/learning-rate) to further tighten the confidence interval on reported metrics.
- Extend authentication to per-user role-based access control (clinician vs. underwriter vs. administrator roles) rather than a single shared login.
- Integrate the pipeline with a larger, continuously updated claims dataset and add automated periodic model retraining.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Practical application of scikit-learn ColumnTransformer pipelines to combine numerical scaling and categorical encoding in a single, reusable, serialisable object.

- Comparative evaluation of linear, tree-based and ensemble regression algorithms using R^2 , MAE, MSE and RMSE on a real-world tabular dataset.
- End-to-end deployment of a trained machine-learning pipeline inside an interactive, authenticated Streamlit web application.

Engineering & Professional Skills Developed

- Structuring an ambiguous, real-world problem into measurable requirements, design alternatives and a trade-off-justified final design (Chapter 3).
- Engineering judgment in interpreting model residuals to identify specific sub-populations requiring human oversight rather than treating a single aggregate metric as sufficient (Section 4.5).
- Team-based software delivery: dividing preprocessing, modelling and deployment responsibilities while integrating the three streams into one working application (Table 4.8).

Individual Reflection

Y. Anjani Priya: The most difficult problem was deciding how to handle the right-skewed charges distribution and the small number of duplicate records without discarding valid data; this was solved by verifying duplicates were exact row matches before dropping them. A key decision was standard-scaling only the numerical features while leaving one-hot-encoded categorical features unscaled, based on scikit-learn documentation recommending this for mixed-type pipelines. Independently, I learned how ColumnTransformer keeps preprocessing reproducible across training and inference. If repeated, I would add automated unit tests for the cleaning step rather than validating it manually.

M. Apurva: The most difficult problem was explaining why the interpretable Linear Regression model under-predicted the highest-cost patients; this was solved by plotting residuals against predicted value (Figure 4.2), which revealed the error was concentrated in the obese-smoker sub-group. A key decision was to benchmark four models rather than tuning only one, based on the evidence in Table 4.4 that ensemble methods clearly outperformed the single linear/tree baselines. I independently learned how Gradient Boosting iteratively fits residuals to capture non-linear interactions that a linear model cannot. If repeated, I would add cross-validation to make the reported R^2 more robust.

Sania Herbert: The most difficult problem was ensuring the dashboard could not be accessed without authentication while keeping the user experience simple; this was solved by gating all navigation behind a session-state login check in app.py. A key decision was to separate the trained-model artefacts from the Streamlit code via joblib serialisation, based on the evidence that this kept prediction latency under the 2-second target (Table 4.3). I independently learned how Streamlit session state persists across reruns of a script. If repeated, I would replace the single shared login with per-user role-based accounts.

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AI-assisted disclosure: Portions of the boilerplate Streamlit UI styling and this report's formatting were drafted with the assistance of an AI writing/coding tool and subsequently reviewed and verified by the student team, in accordance with the Declaration in this report.

APPENDICES

Appendix A – Detailed Calculations

R^2 (Coefficient of Determination): $R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$

MAE (Mean Absolute Error): $MAE = (1/n) \Sigma|y_i - \hat{y}_i|$

MSE (Mean Squared Error): $MSE = (1/n) \Sigma(y_i - \hat{y}_i)^2$

RMSE (Root Mean Squared Error): $RMSE = \sqrt{MSE}$

Appendix B – Engineering Drawings / Schematics

System architecture: insurance.csv → Data Cleaning (drop duplicates) → ColumnTransformer (StandardScaler on age/BMI/children + OneHotEncoder on sex/smoker/region) → 4× Regression Models (Linear Regression, Decision Tree, Random Forest, Gradient Boosting) → Best-model selection by R^2 → joblib serialisation → Streamlit dashboard (Login → Dashboard Home → Module 1 EDA → Module 2 Prediction → Module 3 Evaluation & Decision Support).

Appendix C – Source Code / Code Repository Information

Complete source code is maintained in the submitted project repository (DSA0404_Insurance_Project). Key files: train_models.py (model training/benchmarking), app.py (Streamlit entry point), modules/login.py, modules/dashboard_home.py, modules/module1_preprocessing.py, modules/module2_prediction.py, modules/module3_evaluation.py, and the standalone analysis scripts 01_dataset_inspection.py through 05_insurance_cost_analyzer.py.

Appendix D – Datasheets

Dataset schema (insurance.csv): age (int, years), sex (categorical: male/female), bmi (float, kg/m²), children (int, count), smoker (categorical: yes/no), region (categorical: northeast/northwest/southeast/southwest), charges (float, target variable).

Appendix E – Experimental / Raw Data

Full model comparison, coefficient tables, cost-profile breakdowns and largest-error records are provided as CSV artefacts alongside this report: model_results.csv, module3_regression_coefficients.csv, module3_largest_prediction_errors.csv, module1_statistical_summary.csv, module2_cost_profiles.csv.

Appendix F – Ethics / Institutional Approval

Not applicable — the project uses a fully de-identified, publicly available open dataset (Kaggle Medical Cost Personal Dataset) with no personally identifiable patient information; no separate institutional ethics approval was required.

Appendix G – Project Meeting / Progress Records

Weekly progress milestones are summarised in Table 4.8 (Chapter 4); detailed meeting minutes are maintained in the team's project log.

Appendix H – User / Industry Feedback

Not applicable at this prototype stage — no external clinical or industry pilot feedback was collected.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable — no publications, patents or competition submissions arose from this capstone project.

Appendix J – Individual Contribution Evidence

See the Individual Contribution Statement (page iv) and Table 4.8 (Roles & Actual Contribution) for a detailed breakdown of each student's contribution.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)